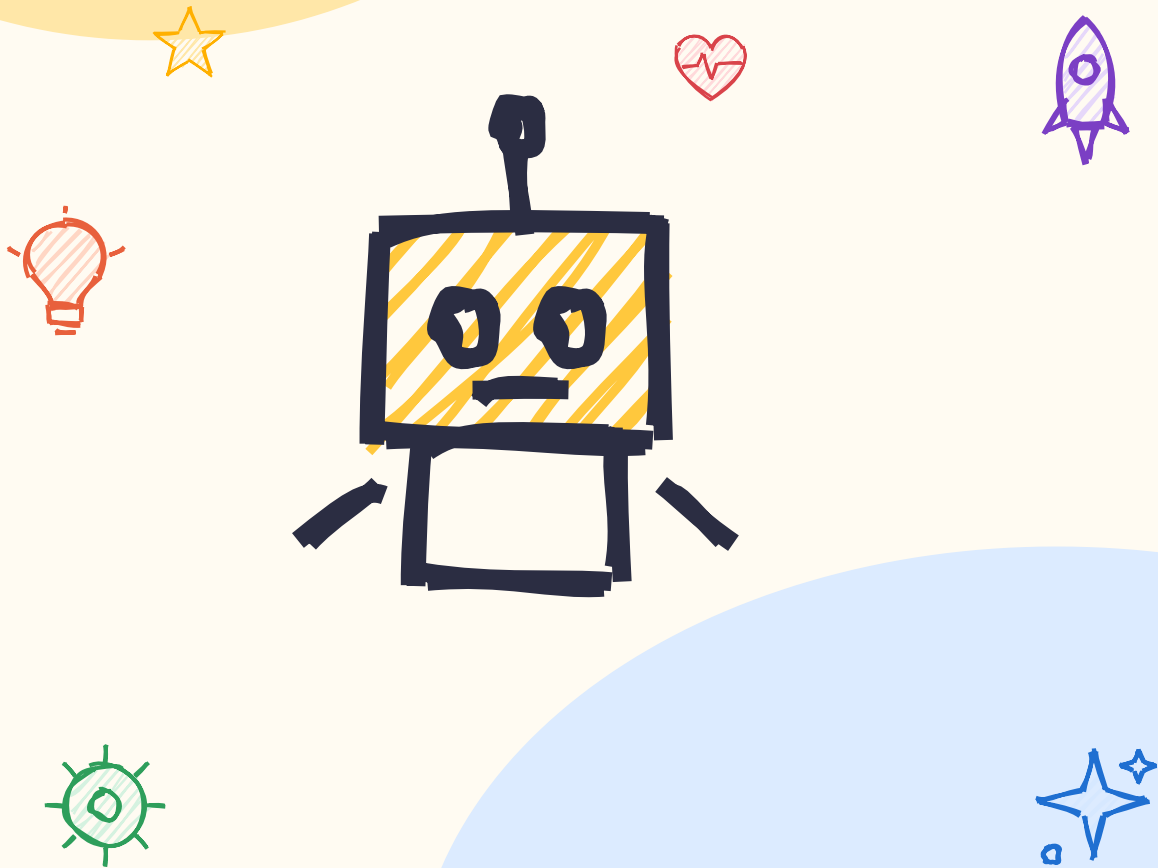


Pak-Turk Maarif presents

Robotics & AI Club

Phase 2 Learning Pathways

The next 16 classes: more coding, new inventions and new challenges for ages 5 to 17.



CURRICULUM DEVELOPED BY

Qasim Mushtaq

MS Robotics and AI

Little Explorers
Ages 5-7

Junior Makers
Ages 8-11

Young Innovators
Ages 12-14

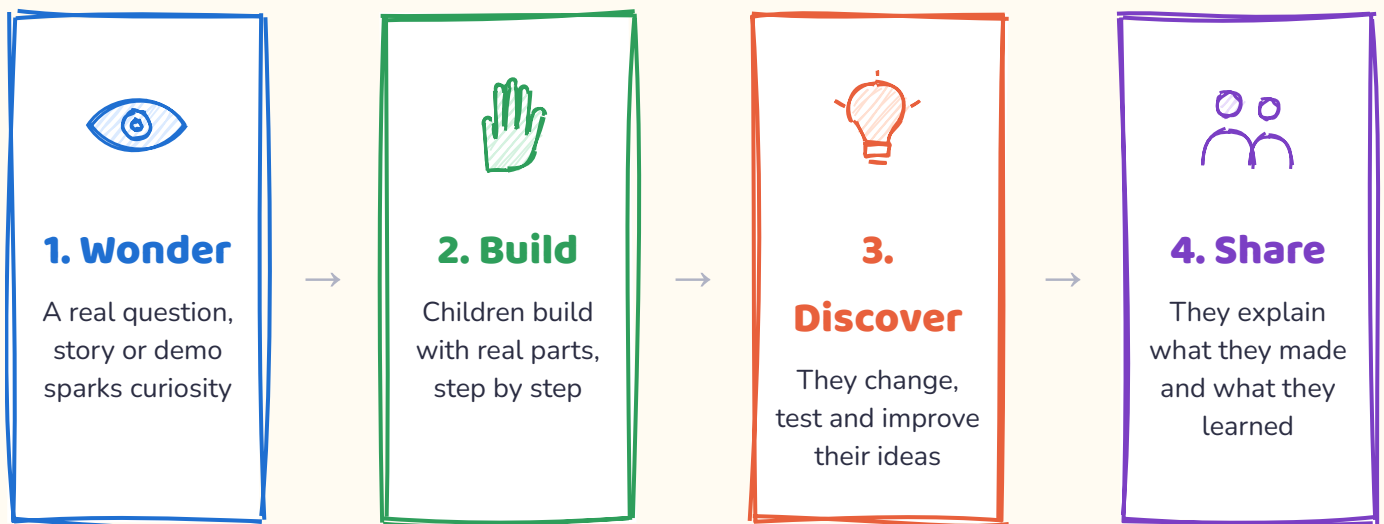
Tech Leaders
Ages 15-17

Dear parents,

Welcome to Phase 2 of the Robotics & AI Club

Technology is shaping the world our children will grow up in. Our club helps them become **creators, not just users**, of that technology. **Phase 2** adds 16 brand-new classes (two a week for 8 weeks). About half of them focus on **coding**, from picture blocks for the youngest to Python, Processing and JavaScript for older students, alongside new hands-on builds. Teams can then design their own project for an end-of-batch showcase.

What happens in a class?



The programme at a glance



Learn by doing

Every class ends with something your child built themselves.



Simulate, then build

Children test ideas safely on screen before using real parts.

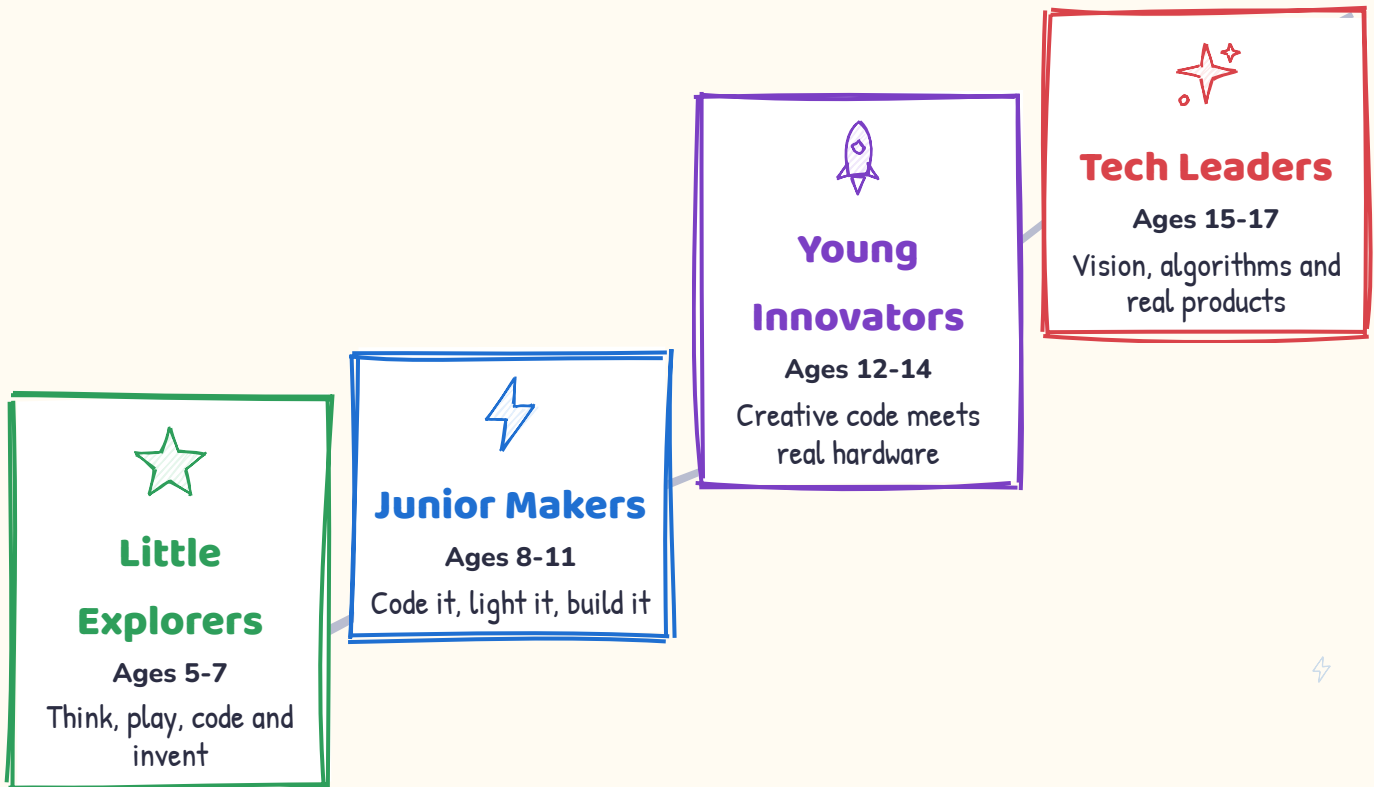


Safe and responsible

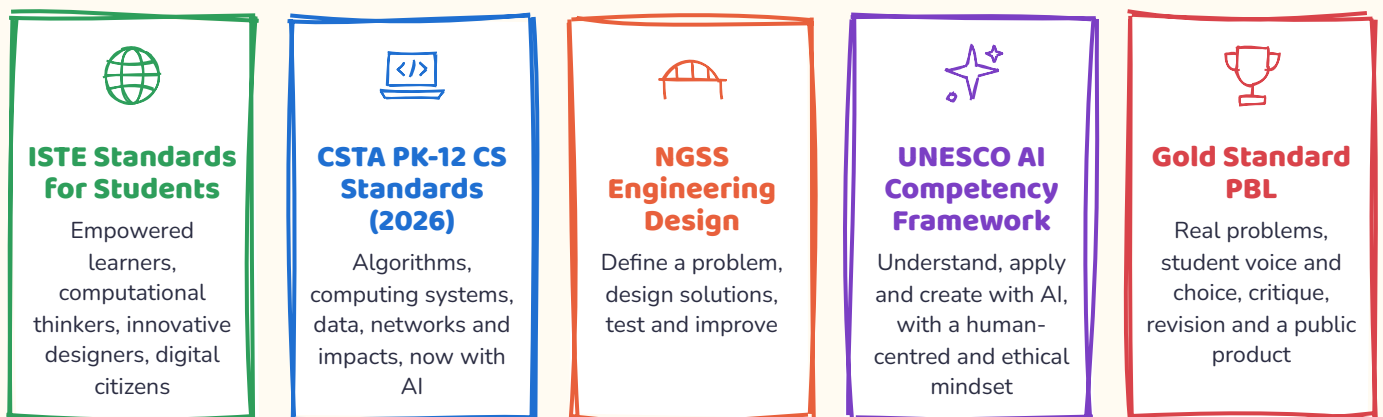
Age-appropriate tools, supervised equipment and a focus on ethics.

Four pathways, one journey

Each age group follows its own Phase 2 pathway: 16 new classes, two a week for 8 weeks, building on what children learned in Phase 1. New students are welcome: a short introduction helps them catch up.



Built on international standards



ISTE: International Society for Technology in Education · CSTA: Computer Science Teachers Association · NGSS: Next Generation Science Standards · PBL: project-based learning

Linked to the UN Global Goals



Every activity carries badges for the UN Sustainable Development Goals it connects to, from clean energy to good health.

Ages 5-7

Little Explorers

Think, play, code and invent



In Phase 2, Little Explorers learn to think like a computer. They give step-by-step instructions to a 'robot chef', solve coding puzzles, program a real robot car through a maze and create dancing characters and talking pets on screen. They light up a rainbow LED ring, make music with buttons, turn movement into electricity and finish by teaching a computer to recognise sounds.

The learning journey



Classes 17-20

Think Like a Computer

- Writes step-by-step instructions and fixes mistakes
- Solves coding puzzles with loops
- Programs a real robot car through a maze



Classes 21-24

Light, Sound & Power

- Mixes colours on a rainbow LED ring
- Plays their own tune with buttons
- Makes light by spinning a motor



Classes 25-28

Stories, Pictures & Machines

- Creates a talking pet and a camera game in Scratch
- Sends a picture as numbers
- Builds a moving cardboard toy



Classes 29-32

Smart Helpers & AI

- Teaches a computer to recognise sounds
- Makes a robot pet follow their hand
- Designs a safety helper robot and launches a rocket



After class 32 • Project showcase!

Teams design and build their own project, then present it at the showcase.

By the end of Phase 2, your child can...



Give clear step-by-step instructions (an algorithm)



Use loops to repeat actions



Program a robot car to follow a route



Create animations and games in ScratchJr and Scratch



Explain how colours of light mix



Explain that machines can make electricity



Explain that computers learn from examples



Design and present an invention that helps people

Skills they grow

Curiosity



Creativity



Problem solving



Coding



Electronics



Teamwork



More dots = more focus in this pathway

Tools they will use

Code.org puzzles

ScratchJr

Scratch

mBlock picture blocks

Robot car

Teachable Machine (teacher-led)

Free, education-friendly tools, used in class with teacher accounts where needed.



Grand finale: project-based learning

A team invention such as a classroom robot helper or an interactive Scratch story with a real button, shown at the showcase. Children practise **planning, teamwork, giving feedback, improving their work and presenting** to an audience.

Try this at home

Play 'Robot Chef' at home: your child gives you exact instructions to make a snack, and you follow them exactly, even the funny mistakes!



Phase 2 turns Junior Makers into coders. They write their first Python programs, design a platform game, create digital art in Processing and explore real data. Alongside the code they build light shows, a musical theremin, a claw machine, a smart traffic junction and a kitchen scale, and they learn to stay safe online in a cyber escape room.

The learning journey



Classes 17–20 Code Studio

- Writes Python code that draws art
- Designs and play-tests a platform game
- Learns binary and secret codes



Classes 21–24 Light, Sound & Music Tech

- Programs colourful LED light shows
- Plays music by waving a hand in the air
- Creates interactive art in Processing



Classes 25–28 Smart City & Mechanisms

- Builds a smart traffic junction and a digital scale
- Builds a joystick-controlled claw machine
- Tests wind turbine designs



Classes 29–32 Code, Data & AI

- Writes a Python guessing game
- Turns class survey answers into charts
- Trains an AI and solves a cyber-safety escape room



After class 32 • Project showcase!

Teams design and build their own project, then present it at the showcase.



By the end of Phase 2, your child can...



Write short Python programs with loops and conditions



Design and test a video game



Explain binary and simple codes



Program LED strips and displays



Combine sensors, motors and code in a working model



Collect data and present it in charts



Train and test a simple AI model



Make safe choices online

Skills they grow

Curiosity



Creativity



Problem solving



Coding



Electronics



Teamwork



More dots = more focus in this pathway

Tools they will use

Python

MakeCode Arcade

Processing

ArduinoBlocks

mBlock 5

CODAP

PictoBlox AI

Free, education-friendly tools, used in class with teacher accounts where needed.



Grand finale: project-based learning

A team project such as a game for younger students or a smart lunchbox scale, presented at the showcase. Children practise **planning, teamwork, giving feedback, improving their work and presenting** to an audience.

Try this at home

Ask your child to teach you the binary code for your initials, or challenge them to guess a number between 1 and 100 in 7 tries.



Young Innovators

Creative code meets real hardware

Young Innovators combine creative coding with real hardware. They create generative art and a live radar screen in Processing, build games in Python and JavaScript, analyse real data and fetch live weather from the internet. They also build a wildlife camera, a posture-correcting wearable, a sumo robot, a walking robot and an energy meter, and use AI to help people communicate.

The learning journey



Classes 17-20

Creative Coding

- Creates generative art and a radar screen in Processing
- Builds a game in Python
- Sets up a Wi-Fi wildlife camera



Classes 21-24

Data & Signals

- Logs and analyses real data in Python
- Builds a posture wearable and sends messages with light
- Writes and breaks secret codes



Classes 25-28

Robots & Mechanisms

- Programs a sumo robot and builds a walking robot
- Measures how much power gadgets use
- Controls hardware from a Python app



Classes 29-32

AI & the Web

- Codes a voice-controlled web game
- Builds a weather dashboard from a live API
- Trains a sign-language AI and builds a custom controller



After class 32 • Project showcase!

Teams design and build their own project, then present it at the showcase.



By the end of Phase 2, your child can...



Write Python programs for games, data and hardware control



Create interactive graphics in Processing and p5.js



Connect computers and microcontrollers over serial



Use web APIs and JSON data



Analyse data and draw conclusions



Explain encryption and code breaking



Design robots and mechanisms that work



Build AI tools that are fair and inclusive

Skills they grow

Curiosity



Creativity



Problem solving



Coding



Electronics



Teamwork



More dots = more focus in this pathway

Tools they will use

Processing

p5.js (JavaScript)

Python + Pygame

Arduino IDE

ESP32-CAM

Teachable Machine

Open weather API

Free, education-friendly tools, used in class with teacher accounts where needed.



Grand finale: project-based learning

A team project such as a study-room data monitor or an accessible game controller, presented with a short pitch. Children practise **planning, teamwork, giving feedback, improving their work and presenting** to an audience.

Try this at home

Ask your child to show you their Processing art or Pygame game, and to explain one line of the code that makes it work.



Tech Leaders tackle the technologies behind self-driving cars, smart grids and modern AI. They build computer vision systems with OpenCV, run machine learning on tiny chips, code flocking and pathfinding algorithms, work in teams with Git, model a solar microgrid and analyse Pakistan's air-quality data. They also design assistive technology, build a neural network from scratch, develop a game in a professional engine and learn how passwords are really protected.

The learning journey



Classes 17–20

Vision & Intelligence

- Builds computer vision with OpenCV
- Steers a robot car using a camera
- Runs AI on a tiny ESP32 chip



Classes 21–24

Algorithms & Systems

- Codes pathfinding and flocking algorithms
- Works like a professional team with Git
- Models a smart solar microgrid



Classes 25–28

Robots & Control

- Programs a robot arm with trigonometry
- Coordinates a swarm of robot cars
- Designs a 3D-printed gearbox and a smart cane



Classes 29–32

AI, Data & Product

- Analyses air-quality data for Pakistan
- Builds a neural network from scratch
- Develops a Godot game and learns password security



After class 32 • Project showcase!

Teams design and build their own project, then present it at the showcase.



By the end of Phase 2, your child can...



Build computer vision programs with OpenCV



Train and deploy AI on microcontrollers (TinyML)



Implement search and simulation algorithms



Collaborate on code with Git and GitHub



Design energy and robotic control systems



Analyse real-world data with pandas



Explain how neural networks learn



Apply ethical cybersecurity practices

Skills they grow

Curiosity



Creativity



Problem solving



Coding



Electronics



Teamwork



More dots = more focus in this pathway

Tools they will use

Python (OpenCV, NumPy, pandas)

Edge Impulse

Processing

Godot Engine

Git + GitHub

Arduino IDE (ESP32)

Onshape / Tinkercad

Free, education-friendly tools, used in class with teacher accounts where needed.



Grand finale: project-based learning

An end-to-end project such as a smog awareness dashboard or an assistive device tested with a real user. Children practise **planning, teamwork, giving feedback, improving their work and presenting** to an audience.

Try this at home

Ask your teen which local problem (smog, traffic, energy) their code could help with, and how they would know if their solution works.

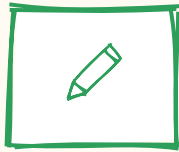
Learning by building real projects

Every pathway ends with two project classes. Our approach follows the internationally recognised **Gold Standard PBL** model by PBLWorks:



A real problem

Projects start with a question that matters



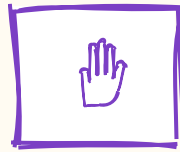
Research & plan

Children ask questions and find answers



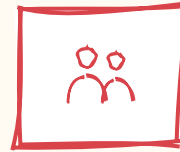
Real world

Real tools, real users, real impact



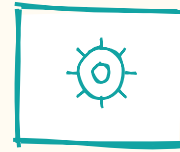
Voice & choice

Teams decide what and how to build



Feedback

Friends and teachers give helpful feedback



Improve

Build, test, fix, repeat



Show the world

A public showcase for families



Safe & responsible technology

- Low-voltage, child-safe electronics
- Hot glue, cutters and soldering handled by teachers for younger children
- AI tools used within their age rules; under-13s use AI only with the teacher
- Lessons on online safety, privacy and the ethics of AI



How you can help

- Ask "What did you build today?" and let them explain
- Celebrate mistakes: debugging is how engineers learn
- Save old boxes, caps and CDs for craft projects
- Come to the end-of-term project showcase



See you in the lab!

Robotics & AI Lab

Pak-Turk Maarif International Schools & Colleges · Islamabad Boys Campus, Chak Shahzad

For questions about the programme, please contact the Robotics & AI Club team through the school office.

Curriculum developed by Qasim Mushtaq, MS Robotics and AI